



Figure 4: A comparison of techniques [Image credit: NVIDIA]

be so precisely described that a machine can be made to simulate it.” It is a broad definition that covers use cases that range from a game-playing bot to a voice recognition system within Siri, as well as converting text to speech and vice versa. It is conventionally thought to have three categories:

- Narrow AI specialised for a specific task
- Artificial general intelligence (AGI) that can simulate human thinking
- Super-intelligent AI, which implies a point where AI surpasses human intelligence entirely

ML is a subset of AI that seems to represent its most successful business use cases. It entails learning from data in order to make informed decisions at a later point, and enables AI to be applied to a broad spectrum of problems. ML allows systems to make their own decisions following a learning process that trains the system towards a goal. A number of tools have emerged that allow a wider audience access to the power of ML algorithms, including Python libraries such as scikit-learn, frameworks such as MLlib for Apache Spark, software such as RapidMiner, and so on.

A further sub-division and subset of AI would be DL, which harnesses the power of deep neural networks in order to train models on large data sets, and make accurate predictions in the fields of image, face and voice recognition, among others. The low trade-off between training time and computation errors makes it a lucrative option for many businesses to switch their core practices to DL or integrate these algorithms into their system.

Classifying applications: There are very fuzzy boundaries that distinguish the applications of AI, ML and DL. However, since there is a demarcation of the scope, it is possible to identify which subset a specific application belongs to. Usually, we classify personal assistants and other forms of bots that aid with specialised tasks, such as playing games, as AI due to their broader nature. These include the applications of search capabilities, filtering and short-listing, voice recognition and text-to-speech conversion bundled into an agent.

Practices that fall into a narrower category such as those involving Big Data analytics and data mining, pattern recognition and the like, are placed under the spectrum of ML



Figure 5: Applications of artificial intelligence [Image credit: The Telegraph, UK]



Figure 6: Deep learning for identifying dogs [Image credit: Datamation]

algorithms. Typically, these involve systems that ‘learn’ from data and apply that learning to a specialised task.

Finally, applications belonging to a niche category, which encompasses a large corpus of text or image-based data utilised to train a model on graphics processing units (GPUs) involve the use of DL algorithms. These often include specialised image and video recognition tasks applied to a broader usage, such as autonomous driving and navigation. **END**

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